

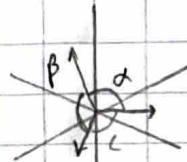
1.2

#130 in my version

+96/100

23. $\cos \beta = \frac{w_1}{\|w\|}, \quad \theta = \beta - \alpha, \quad \cos(\beta - \alpha)$
 $\sin \beta = \frac{w_2}{\|w\|}$

28. $\vec{v} \cdot \vec{w} \Rightarrow$ angle θ formed is 90°



Proof by Geometry & Counts...

As long as there are 3 unit vectors that form 3 angles $> 90^\circ$ with each other.

1.3

4. $w_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, w_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, w_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$

$x_1 w_1 + x_2 w_2 + x_3 w_3 = \vec{0} \quad x_1 = 1$

$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + x_3 \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Let $x = x_2, y = x_3$

$4x + 7y = -1$
 $6x + 9y = -3$

$5x + 8y = -2$
 $6x + 9y = -3$

$4x + 7y = -1$
 $6x + 9y = -3$

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13. $C \vec{x} = \vec{b}$

$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 - x_1 \\ x_4 - x_2 \\ x_5 - x_3 \\ -x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}$

$x_2 = 0 = b_1, -x_4 = 0 = b_5$

$\therefore x_4 = 0 = b_5$

$x_3 - x_1 = x_5 - x_3 = 0 = b_2 = b_4$

$x_3 = 1, x_5 = 1, x_1 = 1$ is

a possible combination

$\therefore \vec{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$

b_1, b_2, b_3 must be zero

$b_1 + b_3 + b_5 = 0$

the meaning is there...

2.1

29. $\vec{v}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
 $A = \begin{bmatrix} 0.8 & 0.3 \\ -0.2 & 0.7 \end{bmatrix}$

$\vec{v}_1 = A \vec{v}_0 = \begin{bmatrix} 0.8 \\ -0.2 \end{bmatrix}$

$\vec{v}_2 = A \vec{v}_1 = \begin{bmatrix} 0.70 \\ -0.30 \end{bmatrix}$

$\vec{v}_3 = A \vec{v}_2 = \begin{bmatrix} 0.45 \\ -0.55 \end{bmatrix}$

The first term appears to be decreasing by a rate that halves by 2, whereas the 2nd term increases by this rate.

30. Using Python:

$u_1 = [0.603125, 0.346875]$

$u_2 = [0.5953125, 0.4046875]$

Converging to $[0.6, 0.4]$

always add up to 1!

$A \vec{s} = \vec{s}$

Letting $\vec{s} = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$

Seems enough...

$A \vec{s} = \vec{s}!$

2.2



20 Row 3 of A is a combination of Row 1 & 2

$$\begin{aligned} x+y+z &= 0 \\ x-2y-2z &= 1 \\ 2x+y+\frac{2}{3}z &= 3 \end{aligned}$$

$$\Rightarrow \begin{aligned} x+y+z &= 0 \\ -2y-2z &= 1 \\ -y-\frac{2}{3}z &= 3 \end{aligned}$$

$$\Rightarrow \begin{aligned} 0z &= \frac{8}{3} \end{aligned}$$

No Solution

32. a. Zero
b. Zero
c.

100 entries

$$A = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & 1 & 1 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & 1 \end{bmatrix}$$

100 entries

d. The row picture would describe a 100^{th} dimensional space passing through the origin. The column picture would describe 100 vectors pointing to $[1, 1, \dots, 1]$ in 100-dimensional space.

2.3

22.



a. $b_3 = \{a_{3j}x_j\}$

$$= a_{31}x_1 + a_{32}x_2 + \dots + a_{3n}x_n$$

d. First component of $E_{21}A_n$ will be the same as b_1

$\Rightarrow \{a_{1j}x_j\} = a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n$

More general!

$E_{21}(A)$ will always have entry $(2,1) = 0$ since it eliminates this entry of A.

Proof by example does not suffice

$E_{21}(E_{21}(A))$ will have an entry $(2,1)$ of $-a_{21}$ since entry $(2,1)$ is eliminated again

29.

Row's Triangle

$$E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \quad P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

After $E_{21}P$, the first 4 rows are that of the I matrix. Then get rid of -1 at E_{21}

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} P = I$$

same similar to P...

$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}$

3.4 $1x=b$

∴ These form the identity matrix

$$\Rightarrow Ax = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow Ax = [I]$$

efficient multiplications $\rightarrow$ I'm a lazy mathematician

$$(AB)C : (2)(4)(10) = 80$$

$$\Rightarrow (2 \times 7)(7 \times 10) = 140$$

$$= 280 \quad 146$$

$$\text{vs. } (4)(7)(10) = 280 \quad 360$$

$$c. \frac{1}{n} + \frac{1}{q} < \frac{1}{m} + \frac{1}{p}$$

$$\Rightarrow (n+q) > (m+p)$$

$$\therefore (ABC) < A(BC)$$

$$(mnp)C < A(npq)$$

$$\frac{mnp + mpq}{mnpq} < \frac{mnp + npq}{mnpq} \Rightarrow \frac{mp(n+q)}{mnpq} < \frac{nq(m+p)}{mnpq}$$

$$= \frac{n+q}{nq} < \frac{m+p}{mp}$$

$$= \frac{1}{nq} + \frac{q}{nq} < \frac{1}{mp} + \frac{p}{mp}$$

$$= \frac{1}{nq} + \frac{1}{n} < \frac{1}{mp} + \frac{1}{m}$$

$$= \frac{1}{nq} + \frac{1}{n} < \frac{1}{mp} + \frac{1}{m}$$

Solution has briefer explanations

2.5

$$4. \{a_{1i} + a_{2i} = a_{3i}\}$$

$$a. A\vec{x} \neq \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

b. anywhere where

$$\vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_1 + b_2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

notation as A_1, A_2

$$\text{or } b_3 = b_1 + b_2$$

$$c. \left[\begin{array}{ccc|c} 1 & 0 & 4 & 7 \\ 2 & 1 & 9 & 2 \\ 3 & 1 & 9 & 9 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 4 & 7 \\ 0 & 1 & -3 & -12 \\ 0 & 1 & -3 & 9 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 4 & 7 \\ 0 & 1 & -3 & -12 \\ 0 & 1 & -3 & 9 \end{array} \right]$$

Wait!

$$= \left[\begin{array}{ccc|c} 1 & 0 & 4 & 7 \\ 0 & 1 & -3 & -12 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Then $(a_{1i} + a_{2i})x_1 + (a_{1i} + a_{2i})x_2 + (a_{1i} + a_{2i})x_3 \neq a_{3i}x_3$

If you distribute

$$0+0 \neq 1$$

In elimination, eqn 3 is reduced to coefficients of zero, or the row vector $\vec{0}$